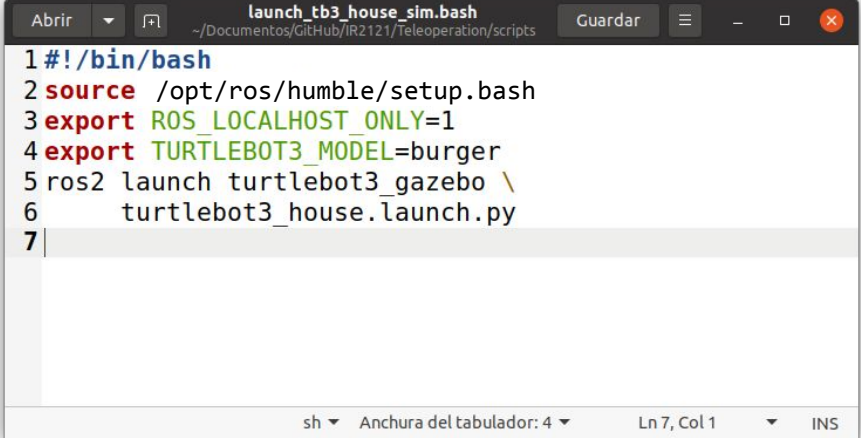


Teleoperation



Teleoperation of TurtleBot 3 in ROS Humble and Gazebo

1. Create a folder named “Teleoperation” in your repository.
2. Create a folder named “scripts” inside the “Teleoperation” folder.
3. Create a BASH script named “launch_tb3_house_sim.bash” with the following lines
4. Change the script properties (permission to execute).
5. Commit and push the repository.
6. Test the script.



The screenshot shows a terminal window titled "launch_tb3_house_sim.bash" with the file path "~/Documentos/GitHub/IR2121/Teleoperation/scripts". The script content is as follows:

```
1 #!/bin/bash
2 source /opt/ros/humble/setup.bash
3 export ROS_LOCALHOST_ONLY=1
4 export TURTLEBOT3_MODEL=burger
5 ros2 launch turtlebot3_gazebo \
6     turtlebot3_house.launch.py
7
```

The terminal window includes standard Ubuntu window controls (Abrir, Guardar, etc.) and a status bar at the bottom showing "sh", "Anchura del tabulador: 4", "Ln 7, Col 1", and "INS".

Teleoperation of TurtleBot 3 in ROS Humble and Gazebo

The screenshot displays a desktop environment with a dark theme. On the left, a vertical dock contains icons for various applications, including a file manager, terminal, and web browser. The main area is divided into three windows:

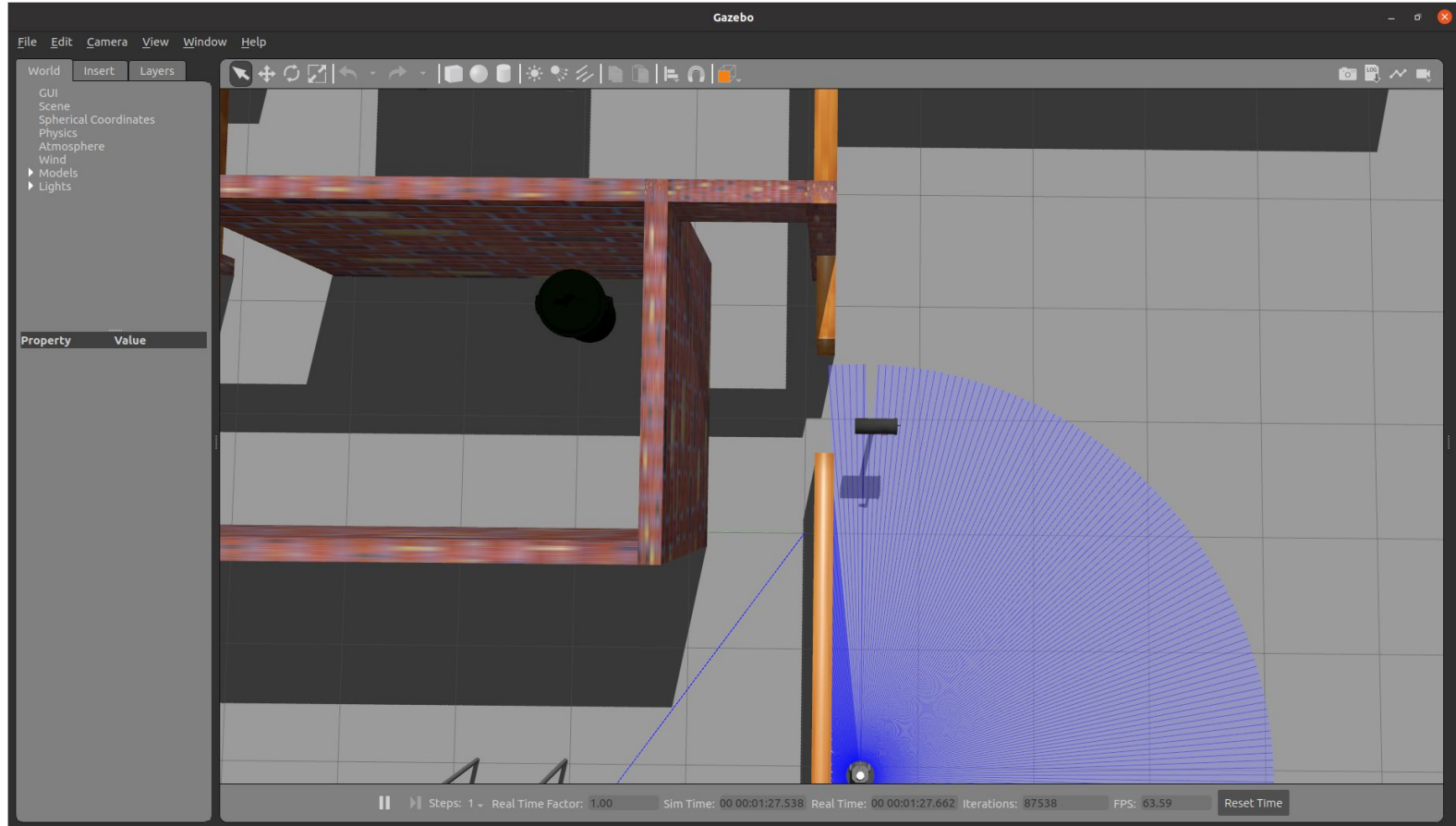
- File Manager (Firefox):** Shows a directory listing for 'Teleoperation' with a file named 'launch_tb3_house_sim.bash' (168 bytes, modified 11:44).
- Terminal (GitHub Desktop):** Displays the 'Create script' dialog for a new repository named 'IR2121'. The 'Current branch' is 'main'. The 'Fetch origin' button is visible. The 'Create script' button is highlighted. Below the dialog, the 'Initial commit' is shown, listing the files: 'Teleop.../launch_tb3_house_sim.bash' and 'turtlebot3_house.launch.py'.
- Web Browser (Firefox):** Displays the GitHub repository page for 'ecervera/IR2121'. The repository is private and has 0 stars, 0 forks, and 1 watching. The 'Code' button is highlighted. The 'README.md' file is shown, containing the text 'IR2121'.

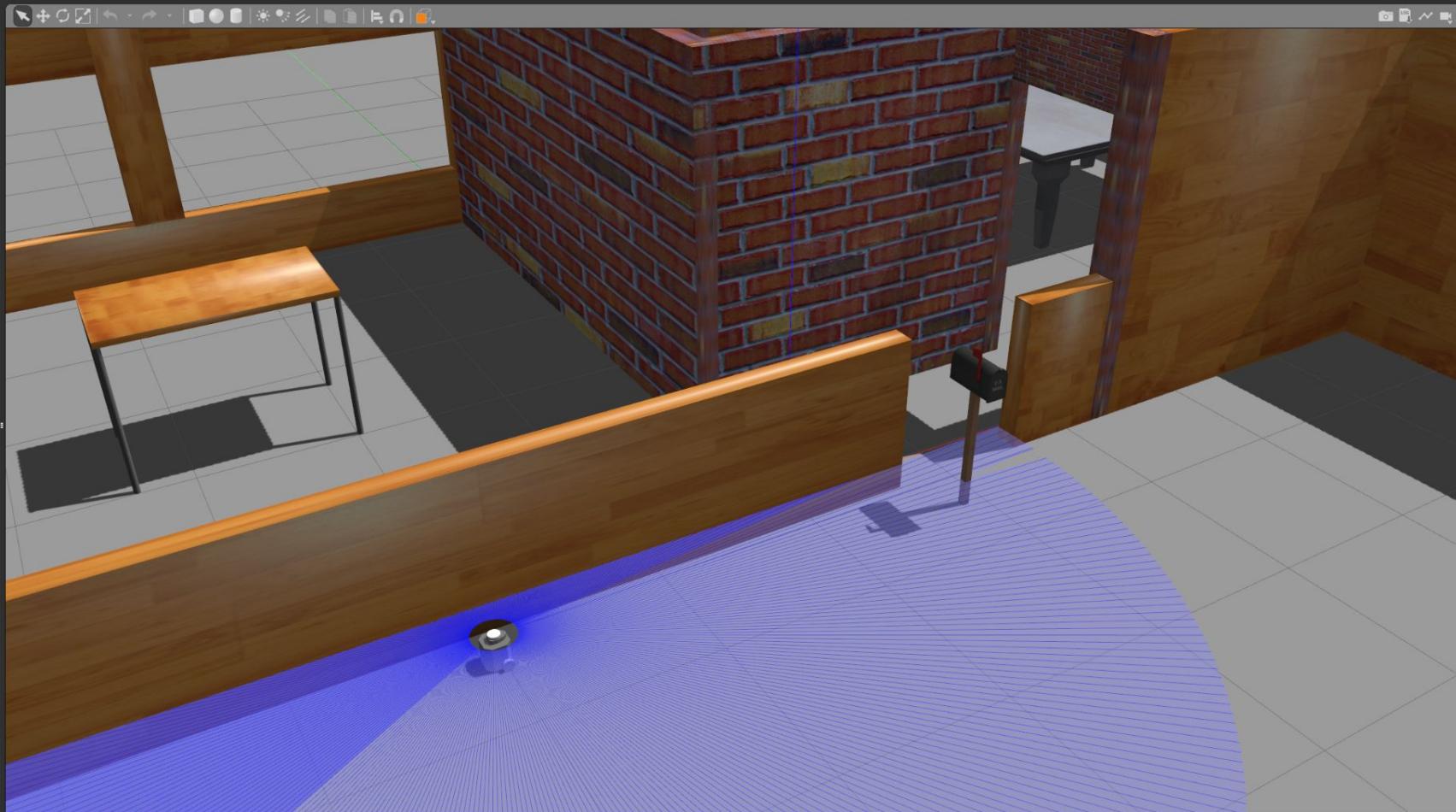
The bottom of the screen shows the system tray with icons for the network, volume, and power. The date and time are displayed as '6 de sep 11:58'.

Teleoperation of TurtleBot 3 in ROS Humble and Gazebo

```
usuario@LABROB06: ~/Documentos/GitHub/IR2121/Teleoperation/scripts
usuario@LABROB06:~/Documentos/GitHub/IR2121/Teleoperation/scripts$ ./launch_tb3_house_sim.bash
[INFO] [launch]: All log files can be found below /home/usuario/.ros/log/2023-09-06-11-52-00-845637-LABROB06-9392
[INFO] [launch]: Default logging verbosity is set to INFO
urdf_file_name : turtlebot3_burger.urdf
[INFO] [gzserver-1]: process started with pid [9394]
[INFO] [gzclient -2]: process started with pid [9397]
[INFO] [robot_state_publisher-3]: process started with pid [9400]
[robot_state_publisher-3] [WARN] [1693993921.175503986] [robot_state_publisher]: No robot_description parameter, but
command-line argument available. Assuming argument is name of URDF file. This backwards compatibility fallback wi
ll be removed in the future.
[robot_state_publisher-3] Parsing robot urdf xml string.
[robot_state_publisher-3] Link base_link had 5 children
[robot_state_publisher-3] Link caster_back_link had 0 children
[robot_state_publisher-3] Link imu_link had 0 children
[robot_state_publisher-3] Link base_scan had 0 children
[robot_state_publisher-3] Link wheel_left_link had 0 children
[robot_state_publisher-3] Link wheel_right_link had 0 children
[robot_state_publisher-3] [INFO] [1693993921.176444524] [robot_state_publisher]: got segment base_footprint
[robot_state_publisher-3] [INFO] [1693993921.176455327] [robot_state_publisher]: got segment base_link
[robot_state_publisher-3] [INFO] [1693993921.176458825] [robot_state_publisher]: got segment base_scan
[robot_state_publisher-3] [INFO] [1693993921.176461749] [robot_state_publisher]: got segment caster_back_link
[robot_state_publisher-3] [INFO] [1693993921.176464663] [robot_state_publisher]: got segment imu_link
[robot_state_publisher-3] [INFO] [1693993921.176467477] [robot_state_publisher]: got segment wheel_left_link
[robot_state_publisher-3] [INFO] [1693993921.176470331] [robot_state_publisher]: got segment wheel_right_link
[gzserver-1] Warning [parser.cc:833] XML Attribute[version] in element[sdf] not defined in SDF, ignoring.
[gzserver-1] [INFO] [1693993922.638846323] [turtlebot3_imu]: <initial_orientation_as_reference> is unset, using defa
ult value of false to comply with REP 145 (world as orientation reference)
[gzserver-1] [INFO] [1693993922.664157108] [turtlebot3_diff_drive]: Wheel pair 1 separation set to [0.160000m]
[gzserver-1] [INFO] [1693993922.664202167] [turtlebot3_diff_drive]: Wheel pair 1 diameter set to [0.066000m]
[gzserver-1] [INFO] [1693993922.664953963] [turtlebot3_diff_drive]: Subscribed to [/cmd_vel]
[gzserver-1] [INFO] [1693993922.666052752] [turtlebot3_diff_drive]: Advertise odometry on [/odom]
[gzserver-1] [INFO] [1693993922.666004940] [turtlebot3_diff_drive]: Publishing odom transforms between [odom] and [b
ase_footprint]
[gzserver-1] [INFO] [1693993922.673107578] [turtlebot3_joint_state]: Going to publish joint [wheel_left_joint]
[gzserver-1] [INFO] [1693993922.673139798] [turtlebot3_joint_state]: Going to publish joint [wheel_right_joint]
```

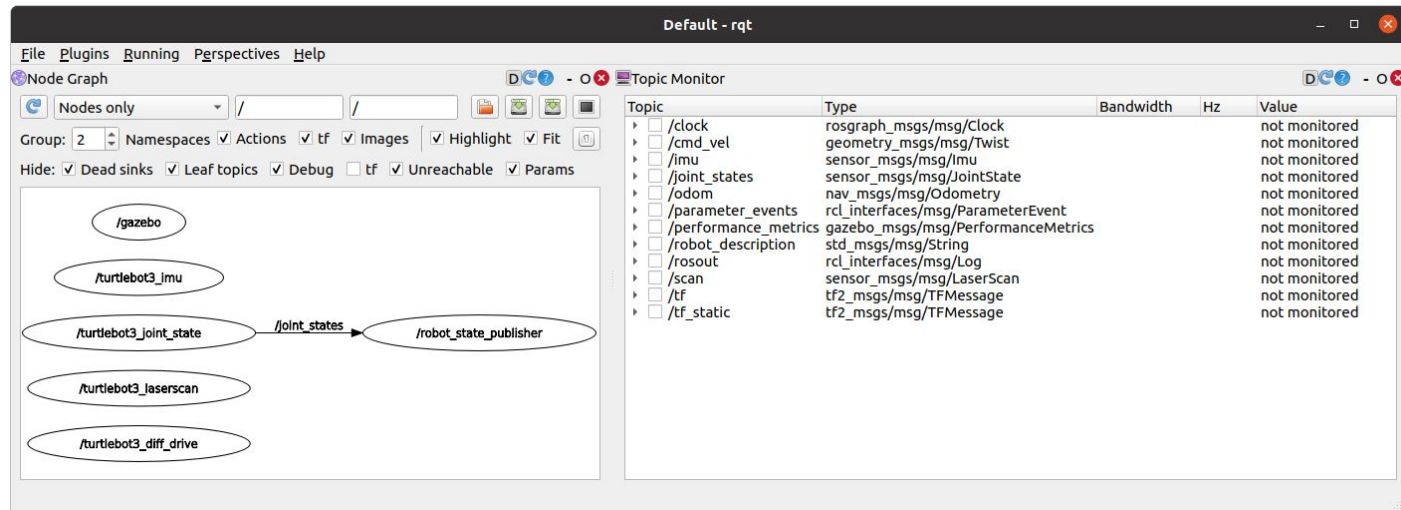
Teleoperation of TurtleBot 3 in ROS Humble and Gazebo



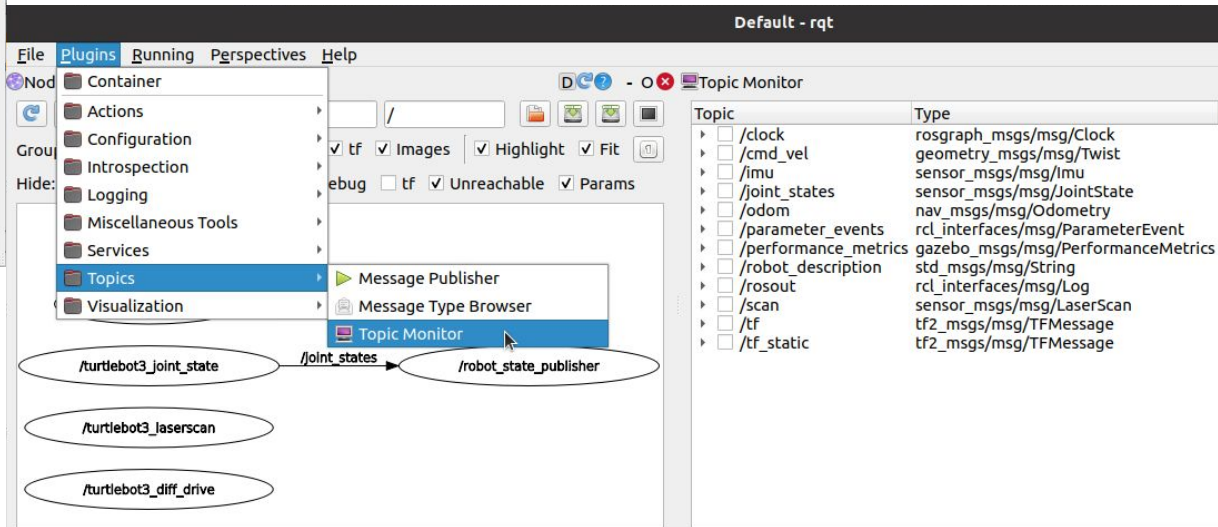
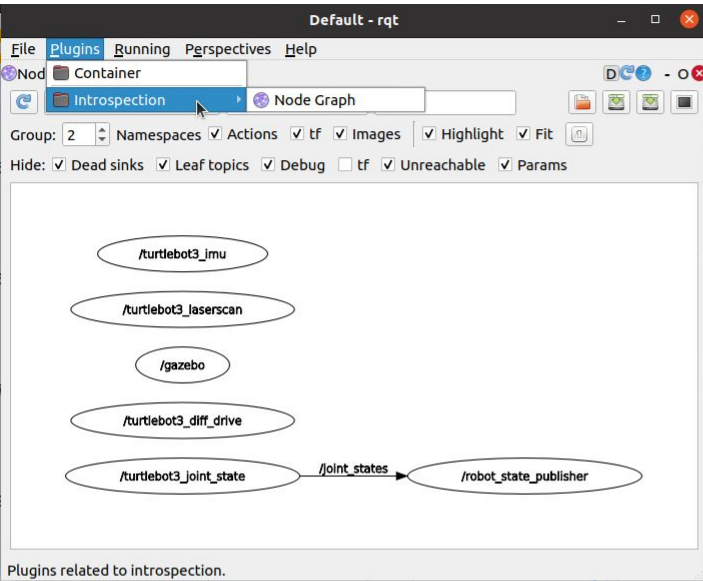


Visualization

1. Create another BASH script named “`launch_rqt.bash`” in the scripts folder.
2. Change the script properties (permission to execute).
3. Commit and push the repository.
4. Test the script.



Visualization

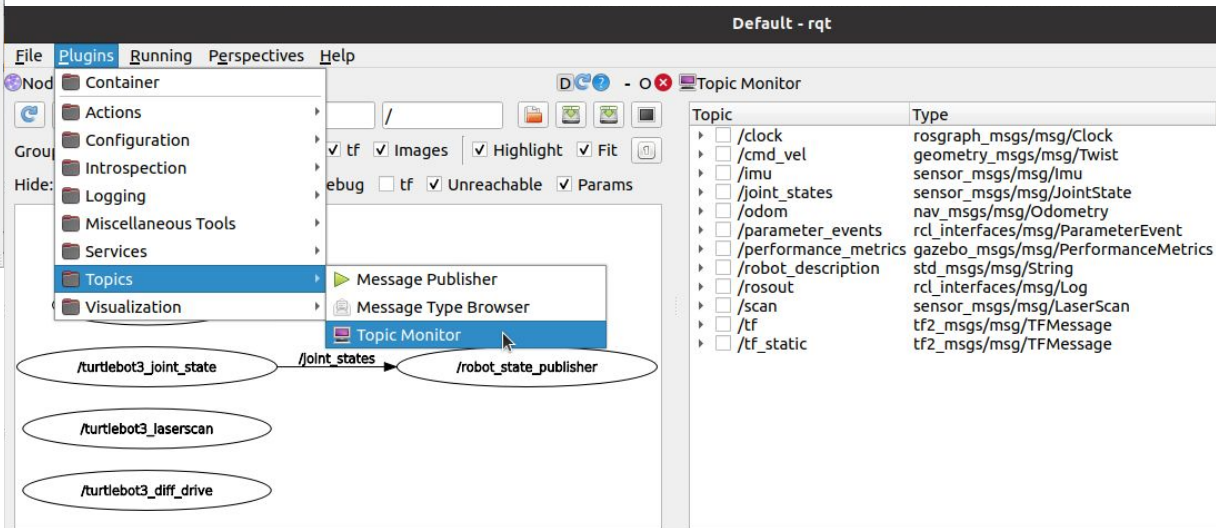
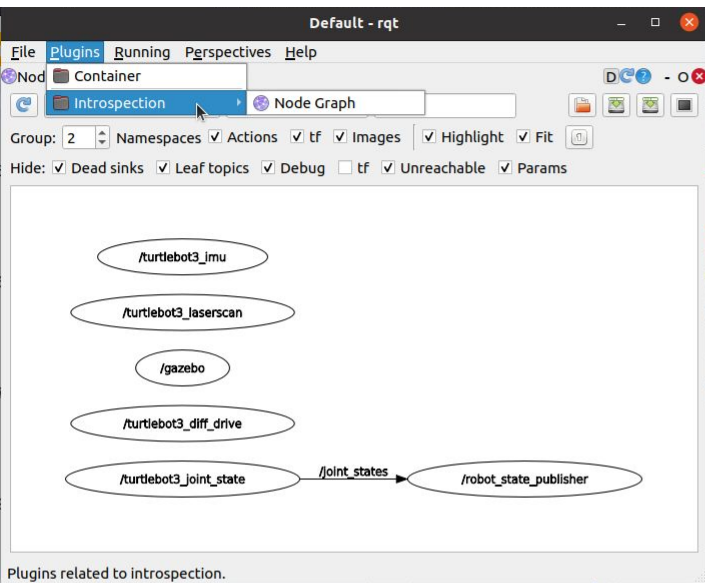


Missing rqt plugins

```
sudo apt update
```

```
sudo apt install \
ros-humble-rqt-common-plugins
```

Launch rqt with **--force-discover** option.



Odometry topic

File Plugins Running Perspectives Help

Node Graph Topic Monitor

Nodes only / /

Group: 2 Namespaces Actions tf Images Highlight Fit

Hide: Dead sinks Leaf topics Debug tf Unreachable Params

/gazebo

/turtlebot3_joint_state → /joint_states → /robot_state_publisher

/turtlebot3_laserscan

/turtlebot3_imu

/turtlebot3_diff_drive

Topic	Type	Bandwidth	Hz	Value
/clock	roscpp_msgs/msg/Clock			not monitored
/cmd_vel	geometry_msgs/msg/Twist			not monitored
/imu	sensor_msgs/msg/Imu			not monitored
/joint_states	sensor_msgs/msg/JointSt...			not monitored
✓ /odom	nav_msgs/msg/Odometry	unknown	28.91	
child_frame_id	string			'base_footpr...
header	std_msgs/Header			
▼ pose	geometry_msgs/PoseWit...			
covariance	double[36]			array([1.e-05,...
				0.e+00, 0....
				0.e+00, 0....
				0.e+00, 0....
				0.e+00, 0....
▼ pose	geometry_msgs/Pose			
► orientation	geometry_msgs/Quaternion			
▼ position	geometry_msgs/Point			
x	double			-1.99989524...
y	double			-0.50000107...
z	double			0.008530375...
twist	geometry_msgs/TwistWit...			
/parameter_events	rcl_interfaces/msg/Param...			not monitored

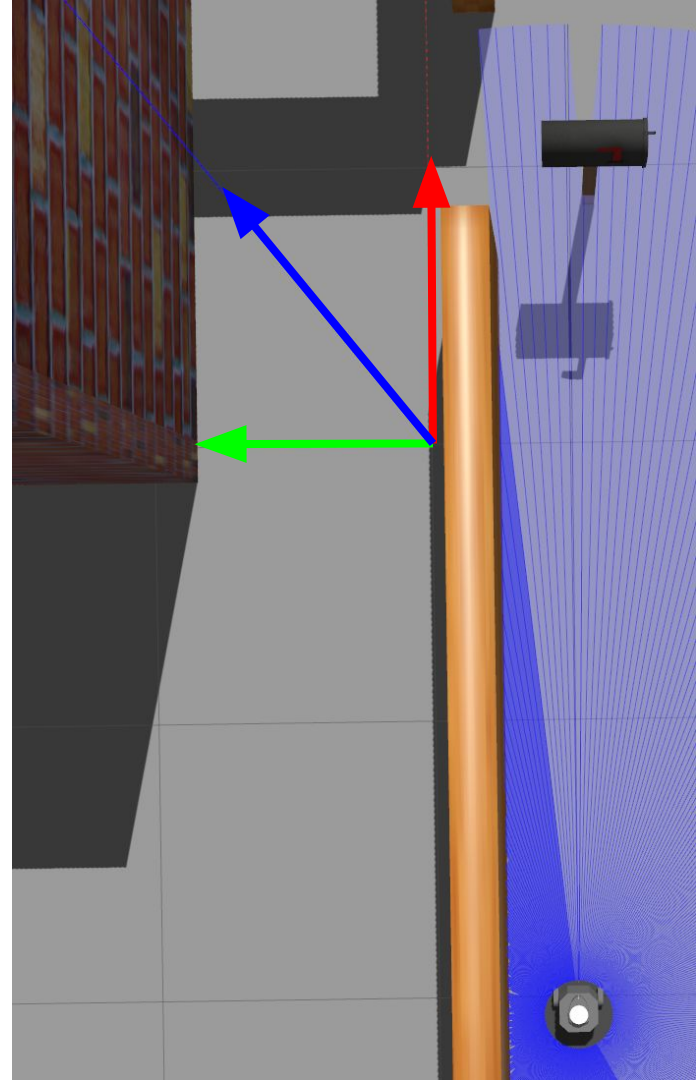
Origin of coordinates

Pose of robot

x -1.9998

y -0.5000

z 0.0085



Teleoperation

1. Create another BASH script named `“launch_tb3_teleop.bash”` in the scripts folder.
2. Change the script properties (permission to execute).
3. Commit and push the repository.
4. Test the script.

Control Your TurtleBot3!

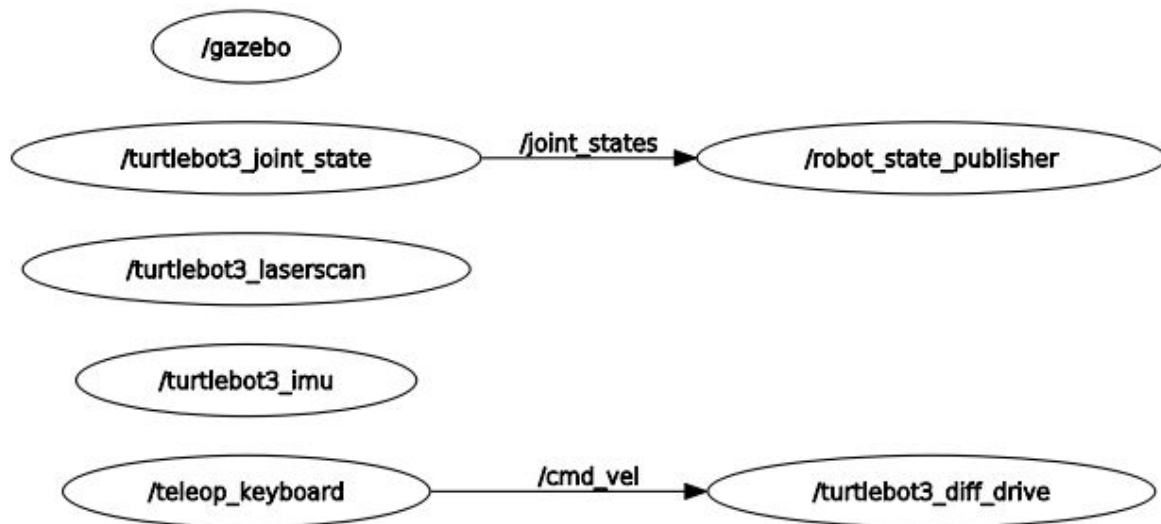
Moving around:

 w
a s d
 x

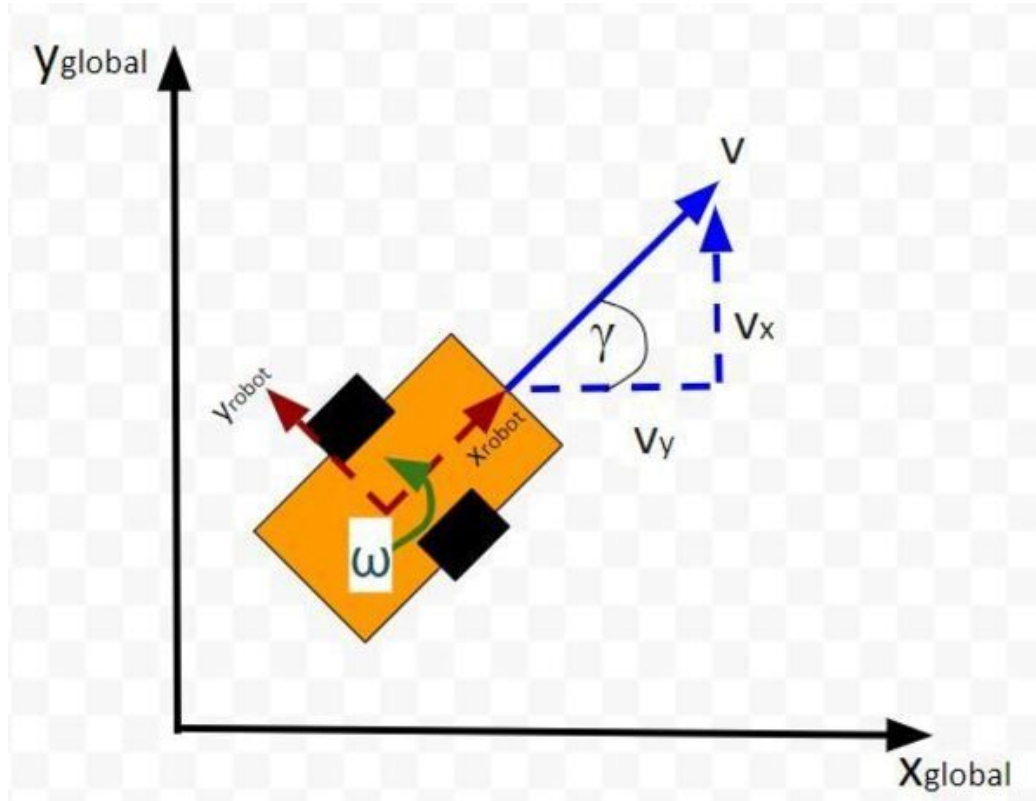
w/x : increase/decrease linear velocity
(Burger : ~ 0.22, Waffle Pi : ~ 0.26)
a/d : increase/decrease angular velocity
(Burger : ~ 2.84, Waffle Pi : ~ 1.82)

space key, s : force stop

CTRL-C to quit



Linear and angular velocities



Task

Teleoperate the robot

Visit all the rooms

Finish at the same initial location

Record topics with `ros2 bag record`

Plot trajectory in RViz

ROS bags

Record

```
source /opt/ros/humble/setup.bash
export ROS_LOCALHOST_ONLY=1
ros2 bag record /clock /odom /tf /tf_static
```

Play

```
ros2 bag play rosbag2_2025_...
```

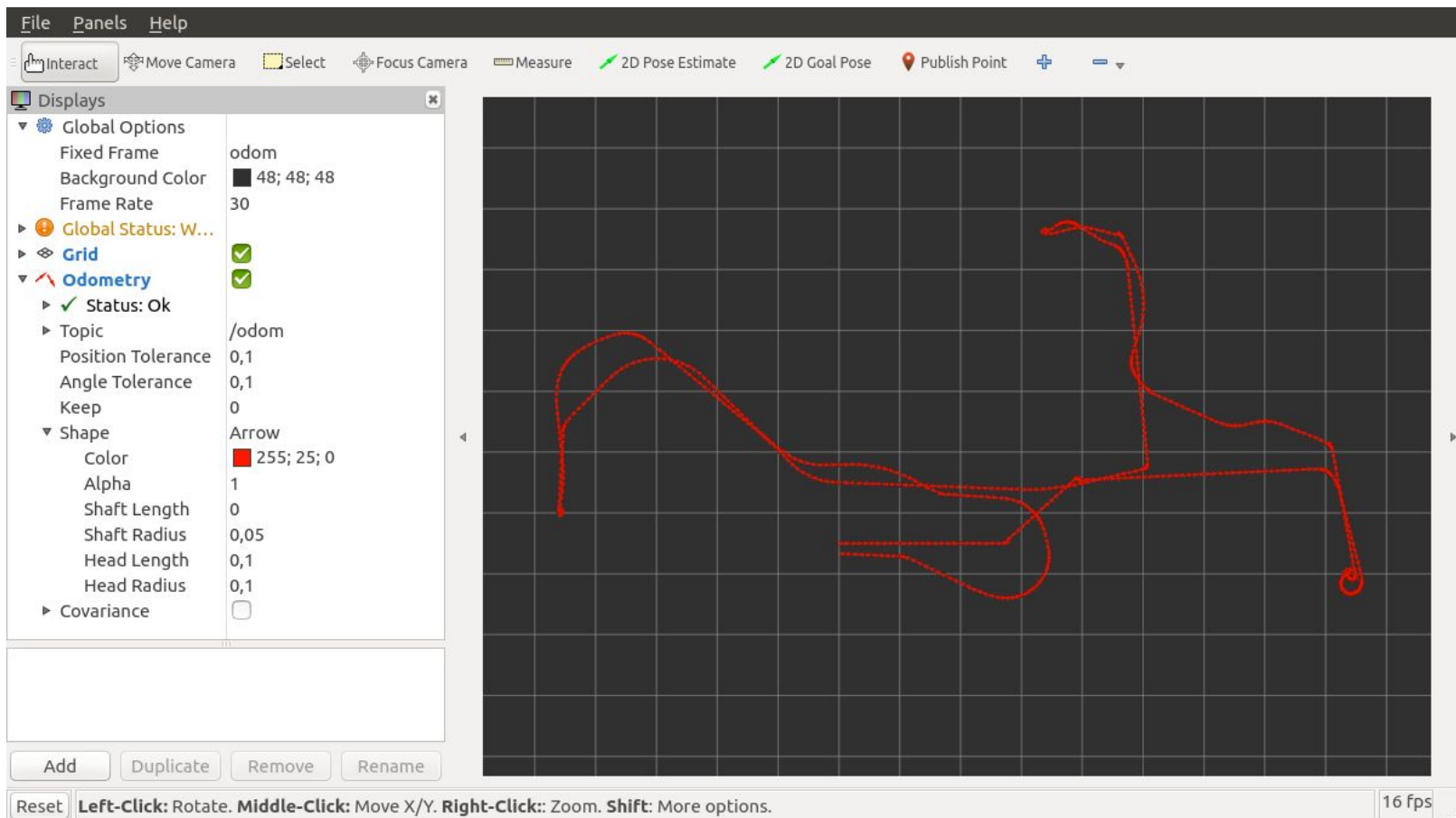
Analyze bag files

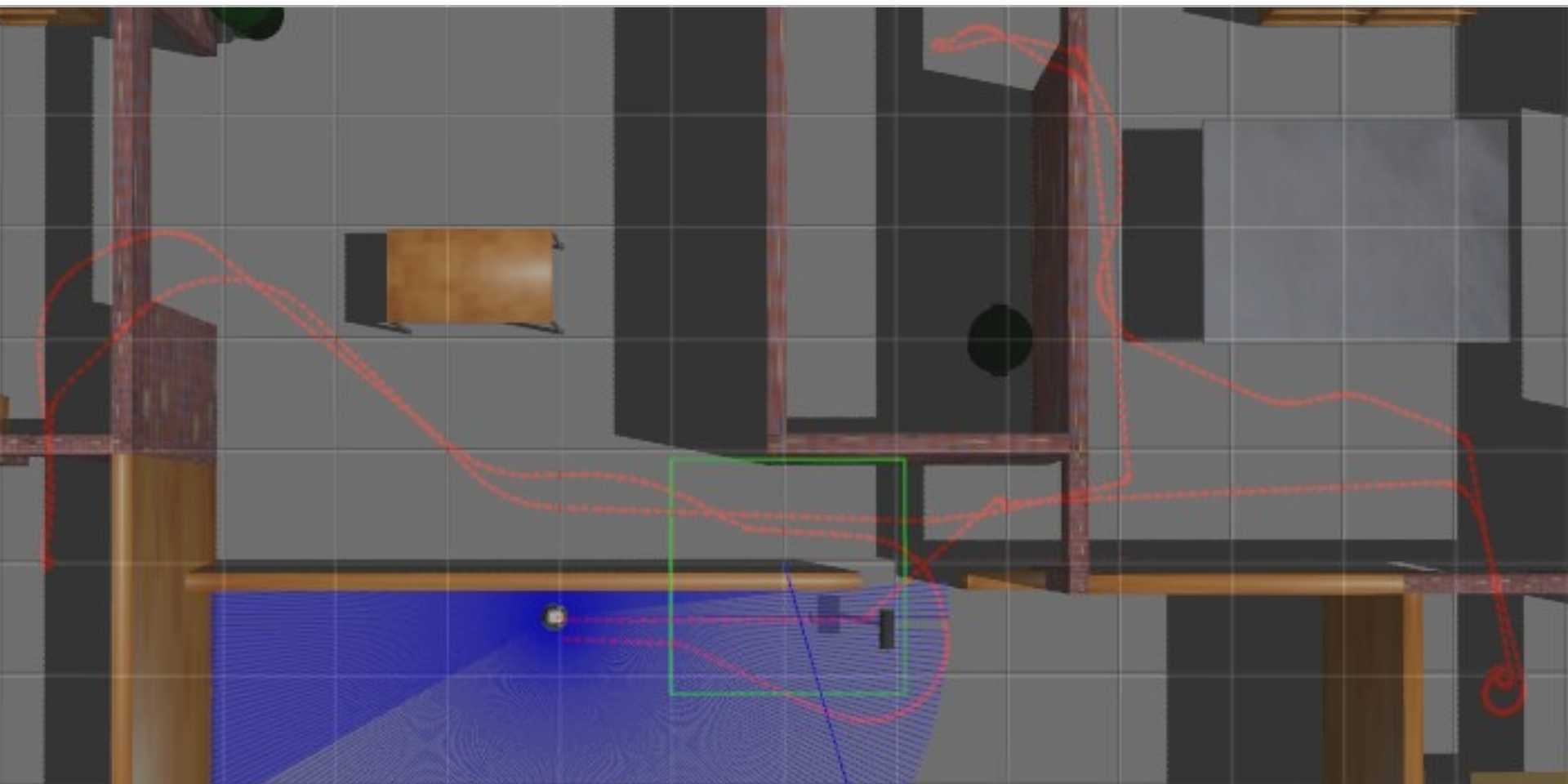
```
ros2 bag info <bagfile>
```

- Bag size
- Duration
- Number of messages
- Topic information:
 - topic
 - type
 - count

RViz

`rviz2 -d config.rviz`





Assignment

Submit to the Aula Virtual the following files:

- ZIP archive of bag folder
- script for launching ros2 bag play
- script for launching rviz
- rviz config file
- picture of the robot's trajectory on top of a simulator image

Recording and Playing

Recording

Gazebo

Teleop

ros2 bag **record** ...

rviz2

Playing

~~Gazebo~~

~~Teleop~~

ros2 bag **play** ...

rviz2